

TEXORA

A Textile Inventory Management System Using Python Django

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INTRODUCTION & PROJECT ABSTRACT

Abstract

Texora is a web-based textile inventory management system developed using the Django framework. It is designed to help textile shops manage their products efficiently by maintaining stock details, tracking quantities, and organizing items based on categories. The system provides a simple and user-friendly interface for monitoring inventory and improving overall business operations. It helps reduce manual effort and enables better control over stock management.

Introduction

Inventory management plays an important role in the smooth functioning of any business. In textile shops, managing different categories of products manually can be time-consuming and may lead to errors. To overcome these problems, Texora provides a digital platform for maintaining inventory records.

The system follows the Model-View-Template (MVT) architecture of Django and allows administrators to add products, manage stock levels, and view inventory details easily. With its responsive interface and organized structure, Texora offers an efficient solution for managing textile inventory.

SYSTEM REQUIREMENTS & DATABASE DESIGN

Software and Hardware Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 500 MB free space

Software Requirements

- Operating System: Ubuntu Linux
- Programming Language: Python 3.x
- Framework: Django
- Database: SQLite
- Frontend Technologies: HTML, CSS, JavaScript
- Development Environment: Visual Studio Code

Database Design

Texora uses Django's Object Relational Mapper (ORM) with SQLite as the backend database. The system stores product information in a table named Product. Each product record contains details required for inventory management.

Model Implementation

The Product model is defined using Django ORM. It stores all information related to textile products and enables efficient management of inventory records. The model ensures that each product has a unique SKU and maintains information such as category, price, quantity, and image details.

Product Table

Field Name	Data Type	Description
S K U	Char Field	Unique product identifier
Name	Char Field	Name of the product
Category	Char Field	Product category (Men's Wear, Women's Wear, Kids Wear)
Subcategory	Char Field	Type of clothing (Formal, Casual, Ethnic, etc.)
Price	Decimal Field	Price of the product
Quantity	Integer Field	Available stock quantity
Image	Char Field	Product image path

SYSTEM ARCHITECTURE & IMPLEMENTATION

System Architecture

Texora follows the Model-View-Template (MVT) architecture of Django. The system consists of three main components: Models, Views, and Templates. The model manages the database, views process user requests and retrieve data, and templates are responsible for displaying information to the user through web pages.

The system enables administrators to manage textile products, monitor stock levels, and maintain inventory records efficiently.

View Implementation

The views act as an intermediary between the database and the user interface. They process requests, retrieve product information, and render the corresponding HTML pages. The system provides

functionalities such as adding products, displaying available stock, and managing inventory details.

Data Visualization

Texora provides a user-friendly interface for displaying inventory information. The dashboard presents product details and stock status in an organized manner, helping users monitor and manage inventory effectively. Graphical representations can also be used to visualize category-wise stock distribution and improve decision-making.

Features Implemented

Dashboard Analytics

The dashboard provides an overview of the inventory by displaying total tracked units, low stock alerts, and the total working capital value. It also includes graphical analysis for category-wise stock distribution.

Product Management

This feature allows users to view all products available in the inventory. Product details such as SKU, category, price, quantity, and stock status are displayed in a tabular format.

Add Product

The system provides a form for adding new products to the inventory. Users can enter product information including SKU, description, category, price, quantity, and image path.

Product Search

A search facility is provided to quickly locate products based on SKU, description, or category, making inventory management easier and more efficient.

Stock Status Monitoring

The system automatically identifies products with low stock and displays warning indicators. This helps prevent shortages and improves inventory control.

Theme Switching

The application supports both light and dark themes. Users can switch between themes to improve readability and user experience.

Image Preview

Each product entry includes an image preview, allowing users to identify products visually and maintain organized records.

Delete Product

Users can remove unwanted or outdated products from the inventory. This feature helps keep the database up-to-date and accurate.

Module Description

Home Module

The Home Module acts as the dashboard of the system. It displays important inventory statistics such as total stock units, low stock alerts and total inventory value. It also provides graphical analysis of category-wise stock distribution.

Functions:

- Display total inventory units.
- Show low stock products.
- Calculate total stock value.
- Display category distribution chart.

Product Management Module

This module manages all available products in the inventory. Users can view product information including SKU, category, quantity, price and stock status.

Functions:

- Display all products.
- Search products.
- Show product images.
- Indicate stock availability.
- Delete products from the database.

Add Product Module

This module allows administrators to insert new products into the inventory database.

Functions:

- Enter SKU code. Enter product details.

- Specify category and subcategory.
- Enter price and quantity.
- Upload image path.
- Store information in database.

About Module

The About Module provides information regarding the purpose and features of the Texora inventory management system.

Functions:

- Describe the system.
- Explain inventory management features.
- Present system capabilities.

Project Files Description

Manage.py: This is the main utility file used to execute Django commands such as running the server and applying migrations.

Settings.py: It contains the configuration settings of the project including installed applications, database settings, templates, and static files.

Urls.py: This file defines the URL patterns and connects web pages to their corresponding view functions.

Models.py: The models file defines the database structure of the application. It contains the Product model used for storing inventory details.

Views.py: This file contains the business logic of the application. It processes user requests and renders the appropriate web pages.

Admin.py: The admin file is used to register models so that they can be managed through Django's admin panel.

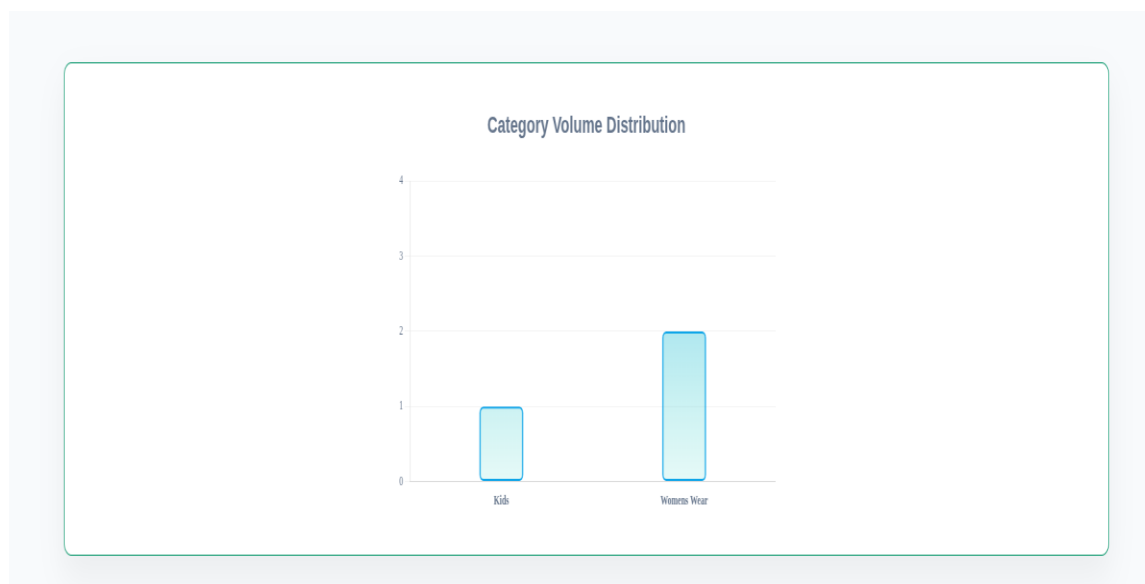
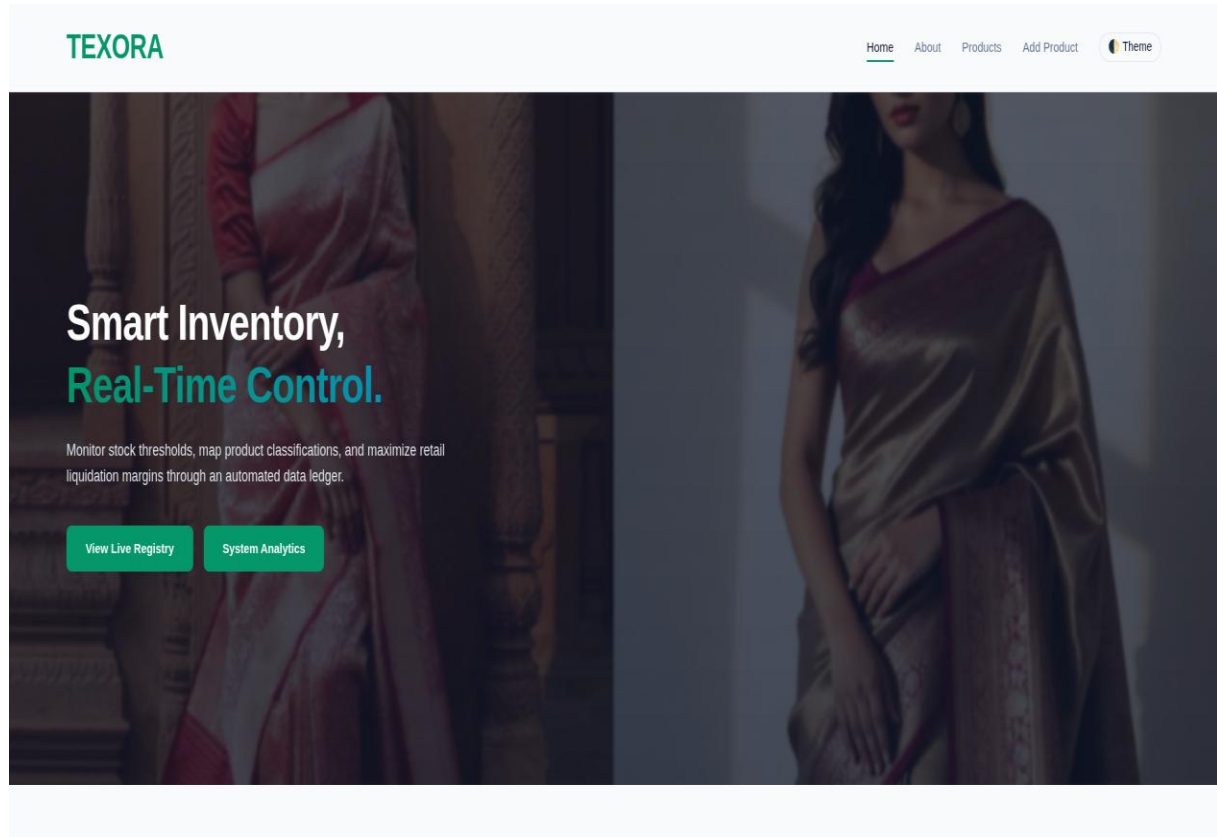
Style.css: This file contains the styling and layout of the web pages and provides a responsive user interface.

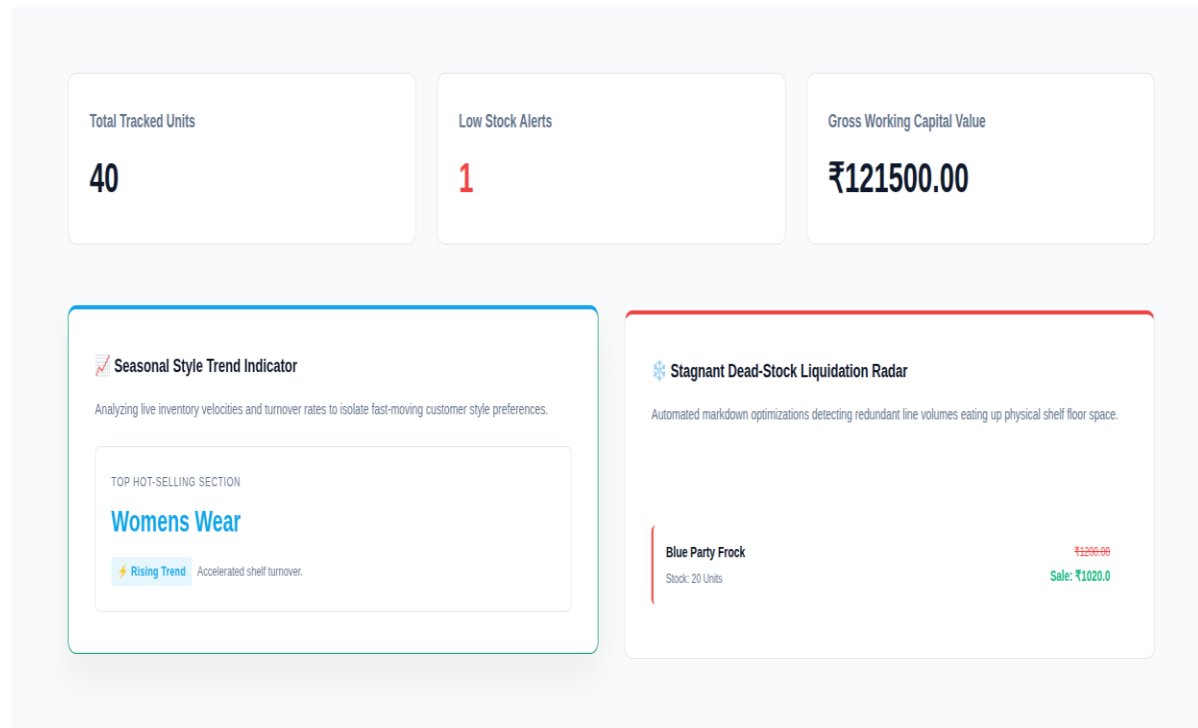
images Folder: This folder stores product images and other graphical assets used in the website.

Db.sqlite3: This is the default SQLite database used to store inventory records and application data.

OUTPUT SCREENS

1) Home Page:





2) About Page:

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HomeAboutProductsAdd ProductTheme

Intelligent Stock Architecture

StockPulse is a high-performance inventory orchestration utility designed to eliminate supply friction, maintain strict asset tracking, and optimize backend warehouse logistics in real-time.

Enterprise Class Architecture

Monitors adjustments, SKU maps, and logistics velocity automatically.

Automated Safety Alerts

Monitors product thresholds instantly and triggers custom color status indicators.

Visual Core Analytics

Aggregates backend data into structured charts for administrative management reviews.

3) Products page:

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Home

About

Products

Add Product

Theme

Live Asset Registry

Search by SKU, Description, or Category...

Visual Preview	SKU Code	Item Description	Classification	Price	Available Stock	Status Metrics	Actions
	#SKU001	Red Banarasi Saree	Womens Wear	₹5000.00	15 Units	Optimal	Delete
	#SKU002	Designer Brown Saree	Womens Wear	₹4500.00	5 Units	Low Stock	Delete
	#SKU003	Blue Party Frock	Kids	₹1200.00	20 Units	Optimal	Delete

4) Add product page:

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Home

About

Products

Add Product

Theme

Boutique Asset Onboarding

Register new garments and luxury products into the main inventory ledger rows.

SKU Code

e.g., TX-SLK-RED-BAN

Garment Description / Model

e.g., Premium Red Banarasi Silk Saree

Main Department (Category)

e.g., Women Wear, Kids Wear

Section Style (Subcategory)

e.g., Saree, Frock, Western

Unit Price (₹)

0.00

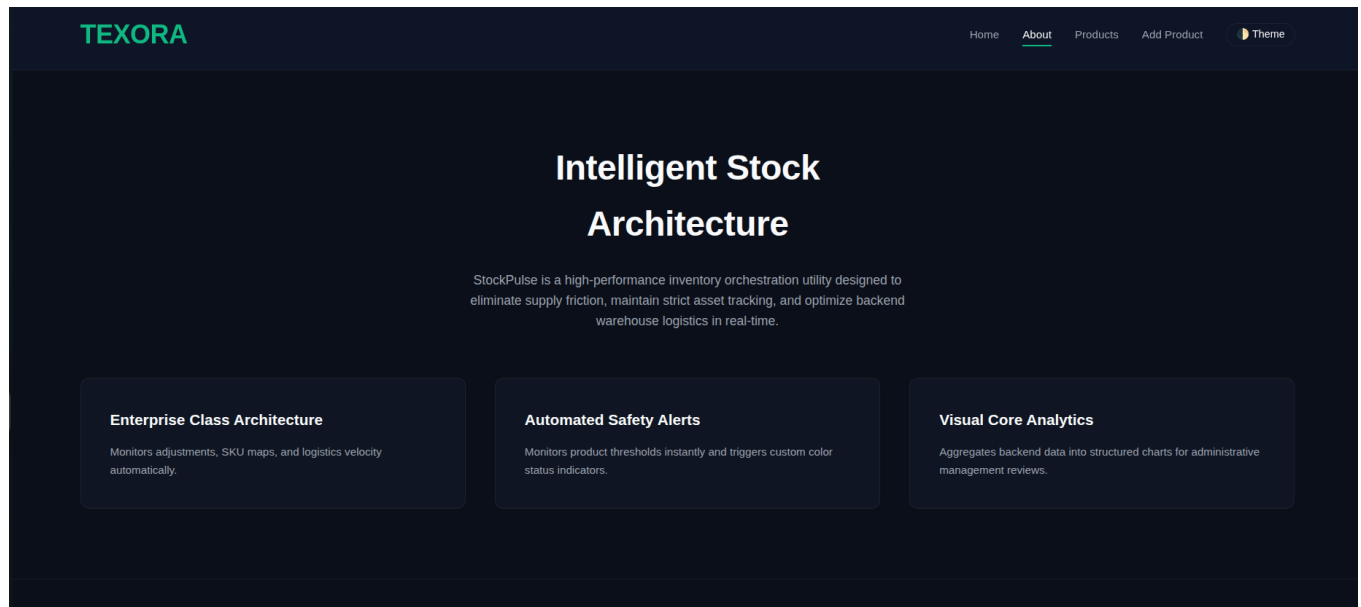
Initial Stock Allocation

0

Showcase Image Local Path

e.g., /static/images/red banarasi saree.jpg

5) Theme change:



CONCLUSION

Conclusion

Texora is a web-based Textile Inventory Management System developed using Python Django. The system provides an efficient and user-friendly platform for managing textile products and monitoring stock levels. It enables users to add products, view inventory details, track stock status, and analyze category-wise distribution through graphical representations.

The project simplifies inventory management by reducing manual effort and improving the accuracy of stock records. With its responsive interface and organized structure,

Texora offers a reliable solution for textile shop inventory management and enhances overall operational efficiency.

Future Scope

The system can be further enhanced by incorporating additional features such as user authentication, AI recommendations, product update and deletion facilities, invoice generation, barcode scanning, sales analysis, and report generation. Integration with cloud databases and advanced analytics can also improve scalability and support larger business operations.